Southeast University

Department of Computer Science and Engineering

Course Name: Database Design Lab

Course Code: CSE384

Lab No: 05

Lab Title: Aggregate Functions and Subqueries

Date of Submission: 29/12/2025

Submitted To:

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Initial: IN

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Course: CSE384

Section: 07

Objective: Learn to retrieve and filter data using basic SELECT statements (AVG, SUM, MIN, MAX, COUNT) and ORDER BY, HAVING.

Problems and Solutions:

Problem 1:

Title: Calculate the average salary of all employees.

SQL Query:
```
SELECT AVG(salary) AS Avg_Salary FROM Employee;
```

Output:

Showing rows 0 - 0 (1 total, Query took 0.0017 seconds.)

```
SELECT AVG(salary) AS Avg_Salary FROM Employee;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

Avg_Salary
55000.0000

Problem 2:

Title:Find the total budget of all departments.

SQL Query:
```
SELECT SUM(budget) AS Total_Budget FROM Department;
```

Output:

Showing rows 0 - 0 (1 total, Query took 0.0010 seconds.)

```
SELECT SUM(budget) AS Total_Budget FROM Department;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

Total_Budget
2150000

Problem 3:

Title: Find the lowest and highest salaries.

SQL Query:
```
SELECT MIN(salary) AS Min_Salary, MAX(salary) AS Max_Salary FROM Employee;
```

Output:

Showing rows 0 - 0 (1 total, Query took 0.0033 seconds.)

```
SELECT MIN(salary) AS Min_Salary, MAX(salary) AS Max_Salary FROM Employee;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

Min_Salary	Max_Salary
45000	65000

Show all | Number of rows: 25 Filter rows: Search this table

Problem 4:

Title: Count the number of employees.

SQL Query:
```
SELECT COUNT(emp_id) AS Emp_Count FROM Employee;
```

Output:

Your SQL query has been executed successfully.

```
SELECT COUNT(emp_id) AS Emp_Count FROM Employee;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

Emp_Count
6

Problem 5:

Title: Find the minimum salary of each department and display only those departments whose minimum salary is 45,000 or more.

SQL Query: `SELECT dept_id, MIN(salary) AS min_salary FROM Employee GROUP BY dept_id HAVING MIN(salary) >= 45000;`

Output:

Showing rows 0 - 2 (3 total, Query took 0.0019 seconds.)

```
SELECT dept_id, MIN(salary) AS min_salary FROM Employee GROUP BY dept_id HAVING MIN(salary) >= 45000;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

	dept_id	min_salary
Edit Copy Delete	1	45000
Edit Copy Delete	2	60000
Edit Copy Delete	3	50000

Check all *With selected:* Edit Copy Delete Export

Problem 6:

Title: Find employees who earn more than the average salary.

SQL Query: `SELECT name, salary FROM Employee WHERE salary > (SELECT AVG(salary) FROM Employee WHERE salary IS NOT NULL);`

Output:

Showing rows 0 - 1 (2 total, Query took 0.0080 seconds.)

```
SELECT name, salary FROM Employee WHERE salary > (SELECT AVG(salary) FROM Employee WHERE salary IS NOT NULL);
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	name	salary
Edit Copy Delete	Nila	60000
Edit Copy Delete	Rina	65000

Check all *With selected:* Edit Copy Delete Export

Problem 7:

Title: Find departments where the maximum salary is above 50000.

SQL Query:
```
SELECT dept_name FROM Department WHERE dept_id IN (SELECT dept_id FROM Employee GROUP BY dept_id HAVING MAX(salary) > 50000);
```

Output:

Showing rows 0 - 1 (2 total, Query took 0.0017 seconds.)

```
SELECT dept_name FROM Department WHERE dept_id IN (SELECT dept_id FROM Employee GROUP BY dept_id HAVING MAX(salary) > 50000);
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

←T→	dept_name
Edit Copy Delete	IT
Edit Copy Delete	Finance

Check all *With selected:* Edit Copy Delete Export

Problem 8:

Title: Find employees with a salary greater than any salary in department 2.

SQL Query:
```
SELECT name, salary FROM Employee WHERE salary > ANY (SELECT salary FROM Employee WHERE dept_id = 2);
```

Output:

Showing rows 0 - 0 (1 total, Query took 0.0016 seconds.)

```
SELECT name, salary FROM Employee WHERE salary > ANY (SELECT salary FROM Employee WHERE dept_id = 2);
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

←T→	name	salary
Edit Copy Delete	Rina	65000

Check all *With selected:* Edit Copy Delete Export

Show all | Number of rows: 25 Filter rows: Search this table

Problem 9:

Title: Find employees with a salary greater than all salaries in department 2.

SQL Query: `SELECT name, salary FROM Employee WHERE salary > ALL (SELECT salary FROM Employee WHERE dept_id = 2);`

Output:

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0005 seconds.)

```
SELECT name, salary FROM Employee WHERE salary > ALL (SELECT salary FROM Employee WHERE dept_id = 2);
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

name	salary

Query results operations

Create view

Problem 10:

Title: Calculate the average salary of employees whose name starts with the first letter of your name.

SQL Query: `SELECT AVG(salary) AS Avg_Salary FROM Employee WHERE name LIKE 'N%';`

Output:

Showing rows 0 - 0 (1 total, Query took 0.0006 seconds.)

```
SELECT AVG(salary) AS Avg_Salary FROM Employee WHERE name LIKE 'N%';
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

Avg_Salary
60000.0000

Problem 11:

Title: Find the total budget of departments where budget is greater than (last 3 digits of your roll × 100).

SQL Query: `SELECT SUM(budget) AS Total_Budget FROM Department WHERE budget > 13000;`

Output:

Showing rows 0 - 0 (1 total, Query took 0.0004 seconds.)

```
SELECT SUM(budget) AS Total_Budget FROM Department WHERE budget > 13000;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

Total_Budget
2150000

Problem 12:

Title: Find the lowest and highest salaries among employees whose salary is greater than (sum of all digits of your roll number).

SQL Query: `SELECT MIN(salary) AS Min_Salary, MAX(salary) AS Max_Salary FROM Employee WHERE salary > 12;`

Output:

Showing rows 0 - 0 (1 total, Query took 0.0005 seconds.)

```
SELECT MIN(salary) AS Min_Salary, MAX(salary) AS Max_Salary FROM Employee WHERE salary > 12;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

Min_Salary	Max_Salary
45000	65000

Problem 13:

Title: Count the number of employees whose name length is equal to the number of letters in your name.

SQL Query: `SELECT COUNT(emp_id) AS Emp_Count FROM Employee WHERE LENGTH(name) = 6;`

Output:

Your SQL query has been executed successfully.

```
SELECT COUNT(emp_id) AS Emp_Count FROM Employee WHERE LENGTH(name) = 6;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

Emp_Count
0

Problem 14:

Title: Find the minimum salary of each department and display only those departments whose minimum salary is ≥ (last 2 digits of your roll × 1000).

SQL Query:
```
SELECT dept_id, MIN(salary) AS min_salary FROM Employee GROUP BY dept_id
HAVING MIN(salary) >= 30000;
```

Output:

Showing rows 0 - 2 (3 total, Query took 0.0005 seconds.)

```
SELECT dept_id, MIN(salary) AS min_salary FROM Employee GROUP BY dept_id HAVING MIN(salary) >= 30000;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

	dept_id	min_salary
Edit Copy Delete	1	45000
Edit Copy Delete	2	60000
Edit Copy Delete	3	50000

Problem 15:

Title: Find employees who earn more than (last digit of your roll × 10000).

SQL Query: `SELECT name, salary FROM Employee WHERE salary > 0;`

Output:

Showing rows 0 - 4 (5 total, Query took 0.0003 seconds.)

```
SELECT name, salary FROM Employee WHERE salary > 0;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	name	salary
Edit Copy Delete	Arif	45000
Edit Copy Delete	Nila	60000
Edit Copy Delete	Sumon	55000
Edit Copy Delete	Rina	65000
Edit Copy Delete	Jamil	50000

Problem 16:

Title: Find departments where the average salary is above (last 3 digits of your roll × 10).

SQL Query: `SELECT dept_name FROM Department WHERE dept_id IN (SELECT dept_id FROM Employee GROUP BY dept_id HAVING AVG(salary) > 1300);`

Output:

Showing rows 0 - 2 (3 total, Query took 0.0005 seconds.)

```
SELECT dept_name FROM Department WHERE dept_id IN (SELECT dept_id FROM Employee GROUP BY dept_id HAVING AVG(salary) > 1300);
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

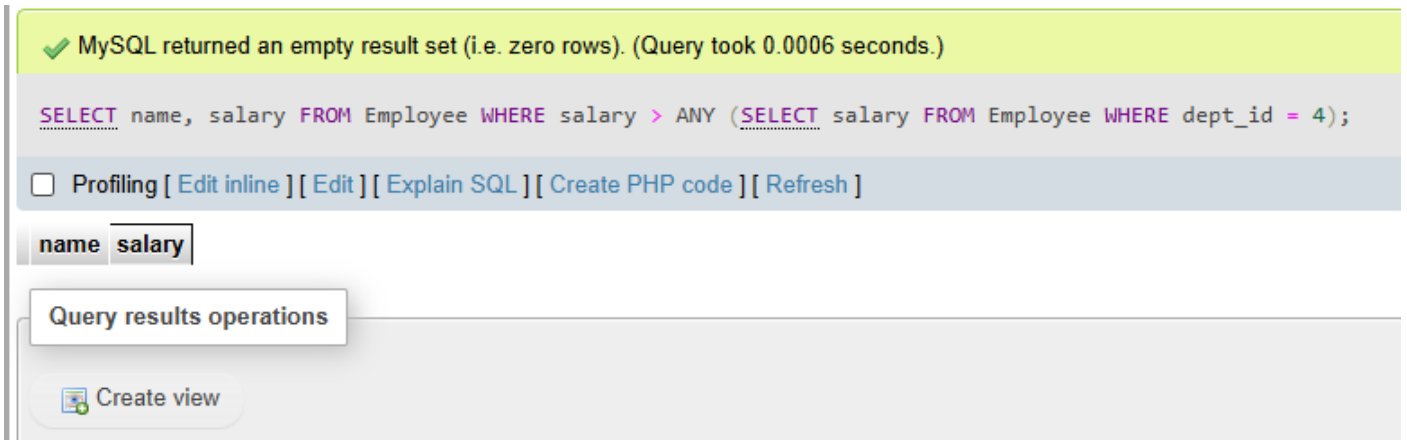
	dept_name
Edit Copy Delete	HR
Edit Copy Delete	IT
Edit Copy Delete	Finance

Problem 17:

Title: Find employees with salary greater than ANY salary in the department equal to your roll's last digit (if last digit > 4 or 0, use 4).

SQL Query:
```
SELECT name, salary FROM Employee WHERE salary > ANY (SELECT salary FROM Employee WHERE dept_id = 4);
```

Output:

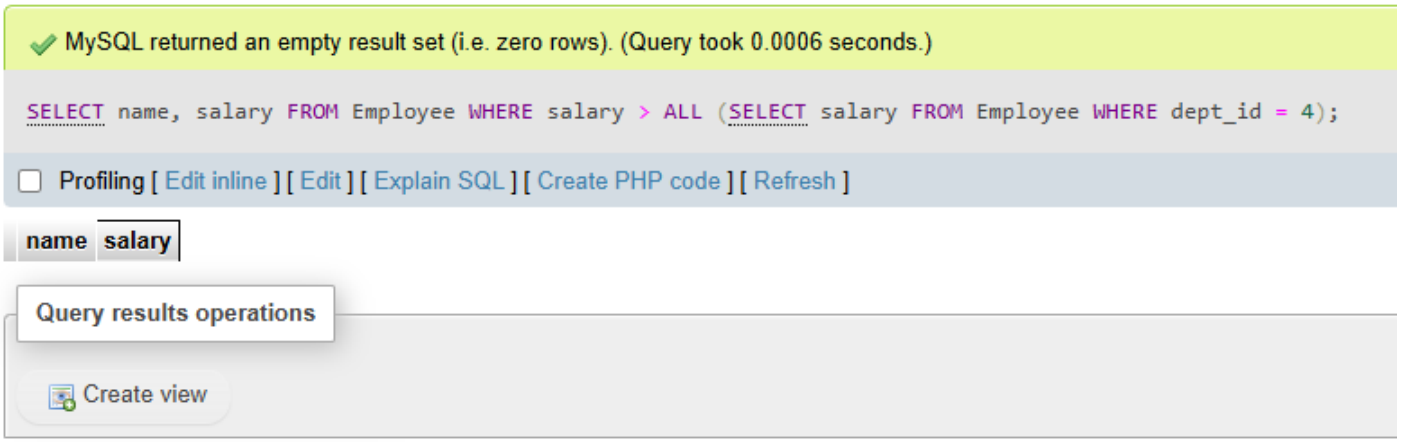


Problem 18:

Title: Find employees with salary greater than ALL salaries in the department equal to the length of your name (if length > 4,use 4).

SQL Query:
```
SELECT name, salary FROM Employee WHERE salary > ALL (SELECT salary FROM Employee WHERE dept_id = 4);
```

Output:



Result: In this lab session, all SQL queries related to aggregate functions and subqueries were successfully executed. Using my name (NAZMUL) and ID (2023100000130) as parameters, I was able to generate specific summary reports from the Employee and Department tables. The results correctly demonstrated how to calculate averages, totals, and counts while filtering grouped data. All nested subqueries (including those using IN, ANY, and ALL) performed as expected, though some returned empty sets due to the NULL values present in the sample data.

Discussion: This lab provided a deeper understanding of how to analyze data rather than just retrieving it. A major observation was the functional difference between the WHERE and HAVING clauses; while WHERE filters individual rows, HAVING was essential for filtering groups after an aggregate function (like MIN or AVG) was applied. Another critical learning point was the behavior of NULL in aggregations. For example, in the case of employee "Karim" (who has a NULL salary), the aggregate functions ignored the record, and subqueries comparing other salaries to his (ANY/ALL) did not yield results. This highlighted the importance of data integrity. Lastly, using subqueries proved to be a powerful way to link the Department table with the results of the Employee table without performing a manual join. 6. Conclusion:

Conclusion: The lab successfully taught me how to summarize large datasets using functions like SUM, AVG, and COUNT. I gained practical experience in categorizing data using GROUP BY and performing complex, multi-layered lookups through subqueries. These skills are vital for generating business reports and performing advanced data analysis. Overall, the exercises solidified my ability to write logic-heavy SQL queries that go beyond simple data selection.